

Lab 03 – Worksheet

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Section: T2

Note: Assumptions and logics should be explained separately in tasks after the task results.

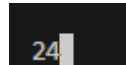
Task 1

Provide appropriately commented codes (Mention question part before each part)

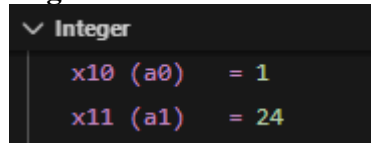
```
addi x10,x0,12
addi x11,x0,12
jal x1,sum #calls the function named sum
addi x11,x10,0
li x10,1
ecall
j exit
sum:
    add x10,x11,x10
    jalr x0,0(x1)
exit:
    j exit
```

Add screenshot of your results (register) & Terminal Console

Ecall result/ Terminal Output:

A terminal window showing the output of the program, which is the number 24.

Register results:

A screenshot of the register results from a debugger. It shows a dropdown menu set to 'Integer' and two registers: x10 (a0) with a value of 1, and x11 (a1) with a value of 24.

Register	Value
x10 (a0)	1
x11 (a1)	24

Task 2

Provide appropriately commented codes (Mention question part before each part)

```
li x10, 1 # g = 1
li x11, 2 # h = 2
li x12, 3 # i = 3
li x13, 4 # j = 4
li x2, 0x100
```

```

jal x1, arithmetic
    addi x11, x20, 0
    li x10, 1
    ecall
j end

arithmetic:
    addi x2, x2, -12
    add x18, x10, x11 # x18 = g + h
    add x19, x12, x13 # x19 = i + j
    sw x18, 0(x2) # store (g + h) to stack
    sw x19, 4(x2) # store (i + j) to stack
    sub x20, x18, x19 # x20 = (g + h) - (i + j)
    sw x20, 8(x2) # store result to stack
    lw x18, 0(x2) # load (g + h) from stack
    lw x19, 4(x2) # load (i + j) from stack
    lw x20, 8(x2) # load result from stack
    addi x2, x2, 12
    jalr x0, 0(x1)

end:
    j end

```

Add screenshot of your results (registers & memory)

Register:

Integer

x00 (zero)	= 0
x01 (ra)	= 24
x02 (sp)	= 256

Integer

x10 (a0)	= 1
x11 (a1)	= -4
x12 (a2)	= 3
x13 (a3)	= 4

Integer

x18 (s2)	= 3
x19 (s3)	= 7
x20 (s4)	= -4

Ecall result/ Terminal Output:

```

-4
Stop program execution!

```

Memory:

0x00000100	0
0x000000FC	-4
0x000000F8	7
0x000000F4	3

Task 3 Provide appropriately commented codes

```
li x10, 0x100 # v[] = 0x100
li x11, 4 # k = 4
li x5, 42 # swap1 = 42
li x6, 84 # swap2 = 84
li x2, 0x200 # stack pointer
jal x1, swap
    li x10, 10
    ecall
j end

swap:
    addi x2, x2, -16
    sw x18, 0(x2)
    sw x19, 4(x2)
    slli x18, x11, 2 # x18 = k * 4 (word size)
    add x18, x10, x18 # x18 = address of v[k] 116
    sw x5, 0(x18) # v[k] = swap1 42
    lw x20, 0(x18) # temp = v[k] 42
    sw x20, 8(x2) # store temp to stack 42
    addi x19, x18, 4 # x19 = address of v[k+1] 120
    sw x6, 0(x19) # v[k+1] = swap2 84
    lw x21, 0(x19) # temp2 = v[k+1] 84
    sw x21, 12(x2) # store temp2 to stack 84
    lw x7, 12(x2) # load temp2 from stack
    lw x8, 8(x2) # load temp from stack
    sw x7, 0(x18) # v[k] = temp2
    sw x8, 0(x19) # v[k+1] = temp
    addi x2, x2, 16 # deallocate stack
    jalr x0, 0(x1)

end:
    j end
```

Add screenshots of your results (registers & memory)

Integer		
x05 (t0)	=	42
x06 (t1)	=	84
x07 (t2)	=	84
x08 (s0)	=	42
x09 (s1)	=	0
x10 (a0)	=	256
x11 (a1)	=	10

Pre-Swapping:

0x00000114	84
0x00000110	42
0x0000010C	0
0x00000108	0
0x00000104	0
0x00000100	0

Post-swapping:

0x00000114	42
0x00000110	84
0x0000010C	0
0x00000108	0
0x00000104	0
0x00000100	0

Task 4 Provide appropriately commented codes

```
li x2, 0x200 # stack pointer
li x10, 0x300 # x10 = address of x
li x11, 0x400 # x11 = address of y

li x5, 72 # 'H'
sb x5, 0(x11)
li x5, 101 # 'e'
sb x5, 1(x11)
li x5, 108 # 'l'
sb x5, 2(x11)
li x5, 108 # 'l'
sb x5, 3(x11)
li x5, 111 # 'o'
sb x5, 4(x11)
li x5, 0 # '\0'
sb x5, 5(x11)

jal x1, strcpy
    li x10, 10
    ecall
j end

strcpy:
    addi x2, x2, -4    # adjust stack
    sw x19, 0(x2)     # save x19
```

```
    li x19, 0          # i = 0

L1:
    add x5, x11, x19    # x5 = address of y[i]
    lb x6, 0(x5)        # x6 = y[i]
    add x7, x10, x19    # x7 = address of x[i]
    sb x6, 0(x7)        # x[i] = y[i]
    beq x6, x0, L2      # if y[i] == '\0' exit loop
    addi x19, x19, 1    # i++
    j L1               # repeat loop

L2:
    lw x19, 0(x2)       # restore x19
    addi x2, x2, 4      # restore stack
    jalr x0, 0(x1)      # return

end:
    j end
```

Add screenshots of your results

0x00000404	o	0x00	0x00	0x00
0x00000400	H	e	l	l
0x00000304	o	0x00	0x00	0x00
0x00000300	H	e	l	l
x00 (zero) = 0x00000000				
x01 (ra) = 0x00000000				
x02 (sp) = 0x00000200				
x03 (gp) = 0x10000000				
x04 (tp) = 0x00000000				
x05 (t0) = 0x00000000				
x06 (t1) = 0x00000000				
x07 (t2) = 0x00000000				
x08 (s0) = 0x00000000				
x09 (s1) = 0x00000000				
x10 (a0) = 0x00000300				
x11 (a1) = 0x00000400				
x12 (a2) = 0x00000000				

Assessment Rubrics

Lab 03: Implementing Jump & Return Instructions in RISC – V Assembly Language

Task No.	LR 2 Code	LR 5 Results
Task 1	/10	/05
Task 2	/10	/10
Task 3	/10	/10
Task 4	/15	/10
Total Points: 100	/45	/35
CLO Mapped	CLO 2	

Affective Domain	Rubric	Points	CLO Mapped
AR7	Report Submission & Git Upload	/10 & /10	CLO2

CLO	Total Points	Points Obtained
2	100	
Total	100	

For description of different levels of the mapped rubrics, please refer to the Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics provided here.

Lab Evaluation Assessment Rubric

#	Assessment Elements	Level 1: Unsatisfactory	Level 2: Developing	Level 3: Good	Level 4: Exemplary
LR2	Program/Code/ Simulation Model/ Network Model	Program/code/simulation model/network model doesnot implement the requiredfunctionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulation model/network model has some errors and doesnot produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation /network model is efficiently implemented and gives correct output. Student has full commandon the basic tools of the software.
LR5	Results & Plots	Figures/ graphs / tables are not developed or are poorlyconstructed with erroneousresults. Titles, captions, units are not mentioned. Data is presented in anobscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are notaccurate. Data presentation is not tooclear.	All figures, graphs, tables are correctly drawn but contain minorerrors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captionsand proper units are mentioned. Data presentation is systematic.
AR9	Report Content/Code Comments	No summary provided. The number/amount of tasks completed below the level of satisfaction and/or submitted late	Couldn't provide good summary of in-lab tasks. Some major tasks were completed but not explained well. Submission on time. Some major plots andfigures provided	Good summary of In-lab tasks. All major tasks completed except few minor ones. The work is supported by some decent explanations, Submission on time, All necessary plots, and figures provided	Outstanding Summary of In-Lab tasks. All task completed and explained well, submitted on time, good presentation of plotsand figure with proper label, titles and captions